

ROBO RUMBLE 2026 · ROBO GRAND PRIX

Lightning McQueen

An autonomous line-following racer that picks its speed from the corner it is about to enter — not the one it is already in.

Team CO Coders

Elimination Round submission

0.86 kg

against a 5 kg limit

300 × 211 mm

against 500 × 500 mm

R 1 843.60

total cost of implementation

The category sets the problem. The constraints set the design.

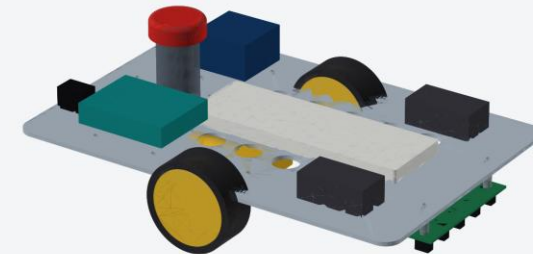
Robo Grand Prix is autonomous racing, so there is no problem to validate — the problem is given: get a robot around a line-marked circuit, on its own, faster than everyone else.

What there **is** to address is a set of hard constraints. A build either satisfies them or it does not.

- **Footprint $\leq 50 \times 50$ cm.** We are at $300 \times 211 \times 107$ mm, and the CAD build script asserts it on every run.
- **Mass below 5 kg.** We are at 0.86 kg — 17 % of the allowance.
- **Accessible ON/OFF switch, and an Emergency Stop.** Both drawn, annotated and placed where an operator can actually reach them.
- **Fully autonomous.** No radio, no Wi-Fi, no Bluetooth anywhere on the vehicle. Any human intervention forfeits the run.

Assembled chassis — isometric

300 × 200 mm acrylic deck · 170 mm track · 65 mm drive wheels · overall envelope 300 × 211 × 107 mm against a 500 × 500 mm limit



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What we are building

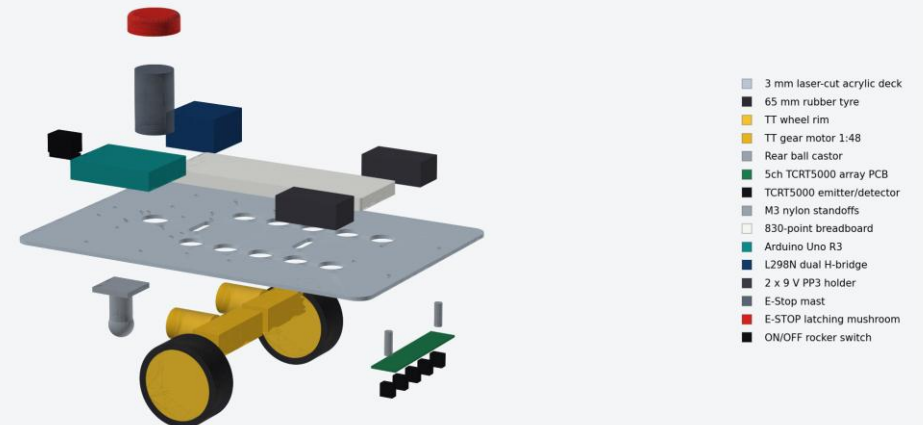
- **Sensing.** Five-channel TCRT5000 reflective array, 20 mm pitch, 12 mm above the floor, mounted 138 mm ahead of the drive axle.
- **Fusion.** A weighted centroid over five per-channel-normalised analog readings gives a cross-track error **in metres** — not a unitless index.
- **Control.** A PID whose output is a **path curvature** in 1/m, running at 200 Hz on an Arduino Uno.
- **Drive.** Two TT gear motors through an L298N H-bridge, from two 9 V PP3 cells on a single switched rail.
- **Structure.** One 3 mm laser-cut acrylic deck, 300 × 200 mm. Every joint bolted; nothing glued.

BUILDABLE, TODAY

Every part is stocked by South African hobby suppliers. The deck is one DXF and ninety seconds on a laser cutter. Nothing here waits on an import order.

Exploded assembly — fastening and stack-up

Deck modules bolt down on M3 nylon standoffs · motors, castor and sensor boom hang below the deck · every joint bolted, none glued



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Our PID instead outputs a **path curvature** κ . A demand of $\kappa = 5 \text{ 1/m}$ is a 200 mm radius turn at any speed. One set of gains, every speed.

- **Speed comes from curvature, before the curvature is applied.** Two ceilings — the point where the outer wheel saturates, and a grip margin — and we take the lower.
- **The sensor bar leads the axle by 138 mm.** That is the distance in which the car sheds speed, so it arrives at the corner already slowed.
- **On saturation, both wheels scale together.** The path curvature survives; only the speed is given up. Clamping one wheel changes the radius and runs you wide.

control loop, 15 % CPU

analog channels — a continuous error, so the derivative term is usable

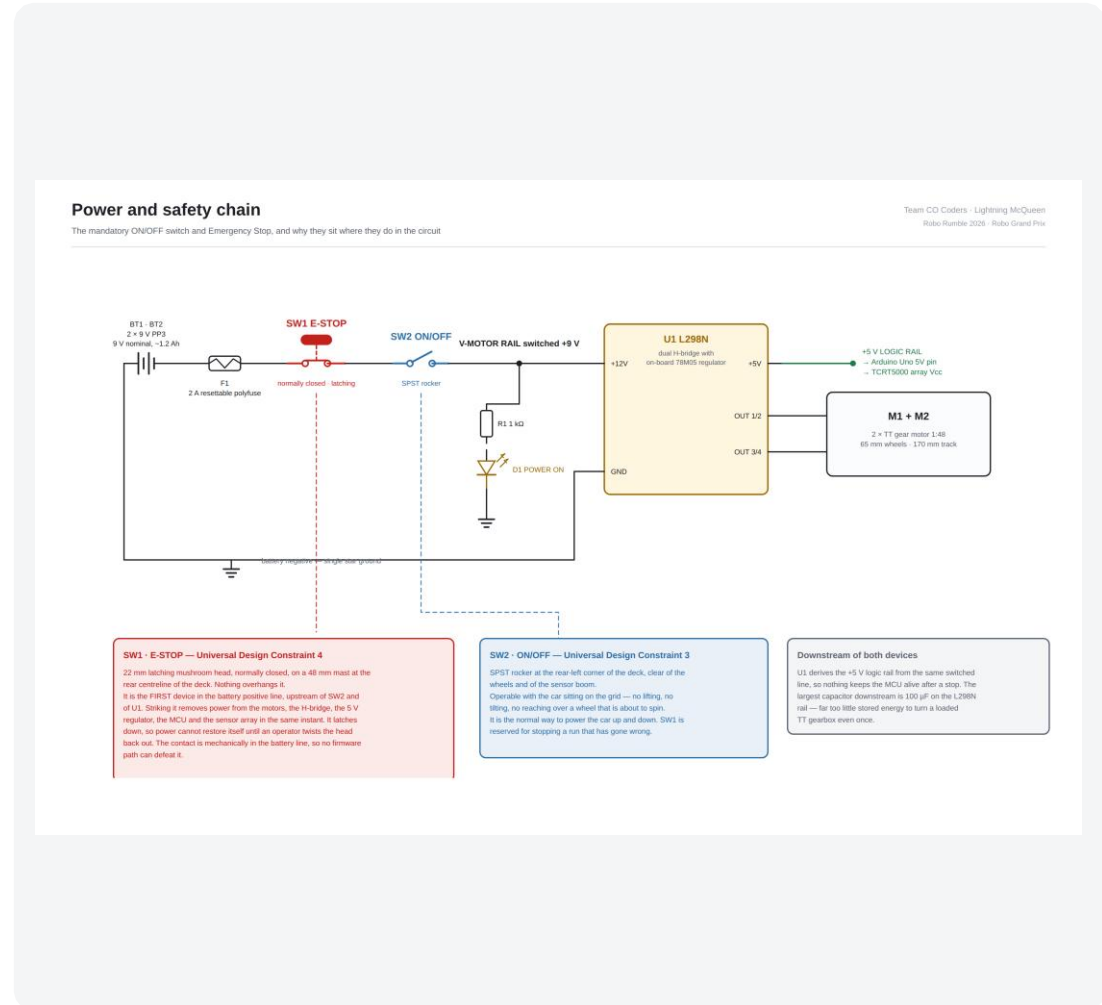


The E-Stop is the first device in the battery line

- **SW1 — E-Stop.** 22 mm latching mushroom head, normally closed, on a 48 mm mast at the rear centreline. Tallest object on the car, nothing overhanging it.
- **SW2 — ON/OFF.** SPST rocker at the rear-left corner, operable with the car on the grid — no lifting, no reaching over a wheel.
- **Everything is downstream.** The L298N's 78M05 derives the 5 V logic rail from the same switched line, so one strike de-energises the motors, the H-bridge, the MCU and the sensors together.
- **Firmware cannot defeat it.** The contact is mechanically in series with the battery. Nothing is left running once it opens.
- **Stored energy cannot move the car.** 100 μF at 5 V is 1.25 mJ; turning a loaded TT gearbox needs about a joule.

WHY THERE IS NO SEPARATE LOGIC BATTERY

A second supply would keep the MCU alive after the motor rail opened — exactly what Constraint 4 exists to prevent. One rail, one stop.



05 · COST OF IMPLEMENTATION

R 1 843.60, all in

CATEGORY	COST	SHARE
Control & drive electronics	R 480.00	28.6 %
Chassis & hardware	R 440.00	26.3 %
Power & safety	R 301.00	18.0 %
Interconnect	R 230.00	13.7 %
Sensing	R 167.00	10.0 %
Indication & passives	R 58.00	3.5 %
Contingency – spares and breakage (10 %)	R 167.60	—

TOTAL COST OF IMPLEMENTATION R 1 843.60

24 line items, 28 units. Full breakdown with quantities, suppliers, stock codes and buy links in [Bill_of_Materials.xlsx](#).

0.86 kg

assembled mass — 17 % of the 5 kg allowance. Two 9 V PP3 cells weigh 92 g where twelve AA cells weigh 276 g.

R 145

the E-Stop – the most expensive single item, and the one line we will not economise on.

R 180

laser cutting the deck. Buys dimensional accuracy we cannot get by hand, and a re-cut takes ninety seconds.

Team CO Coders

[Christopher Lukanga]

Team lead · Controls

Control law, firmware architecture, Webots simulation and the speed schedule.

[Olivier Nzamba]

Electronics

Component selection, schematics, power and safety chain, wiring loom.